

Project Tech Stack Report

LMS & Website Redesign

Govt. Graduate College for Women Township Lahore

<https://ggcw.edu.pk/>

Project Overview

This project involves redesigning the official website of Govt. Graduate College for Women Township Lahore and developing a modern **Learning Management System (LMS)**. The goal is to improve digital learning, user experience, and administrative efficiency using a scalable full-stack solution.

Objectives

- Modern UI/UX redesign of existing website
 - Develop a complete LMS for students, teachers, and admin
 - Enable online learning, course management, and progress tracking
 - Improve performance, security, and scalability
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Tech Stack

Frontend

- Next.js (React Framework for SSR & performance)
- Tailwind CSS (modern responsive UI)
- Axios / Fetch API (API communication)

Backend

- Node.js
- Express.js
- RESTful APIs
- JWT Authentication

Database

- MongoDB
- Mongoose (ODM)